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Essay / Assignment Title: Design, Analysis, and Optimization of a Relational Database for an E-Commerce Order Management System

Programme title: Enterprise Data Warehouses and Database Management Systems

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Abstract

Through the presentation of a relational database for an E-Commerce Order Management System, this paper showcases the dependence of real business operations on the theoretical database principles. The project covers the entire database development life cycle, starting from domain analysis and design of the normalized schema to Third Normal Form, and going up to filling with sample data and a using advanced SQL for analysis. The use of concepts like views, window functions, stored procedures, and user-defined functions is paramount in providing meaningful business analysis. Moreover, the task analyses the advantages and disadvantages of distributed databases by applying the CAP Theorem and considers a number of query optimization methods that will eventually result in higher performance and scalability in data-intensive e-commerce systems.

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Introduction

One of the key benefits of modern organizations is the fact that they rely heavily on data-driven decision-making and this in turn has made efficient database design and management a technical and strategic capability that is critically essential. The purpose of the current assignment is to perform a detailed study of relational database concepts through the selected e-commerce order management system as a real-world application. The entire database development life cycle is being displayed starting from domain analysis and normalized schema design to data population, analytical query development, distributed system considerations, and performance optimization. The e-commerce domain is selected not only for its complexity but also for the high transaction volume and the necessity of the domain for modern digital businesses. Data models and optimized queries are the means that must be utilized in customer, product, order, payment, and supplier management as well as analytics in order to keep accuracy, scalability, and reliability. Students will benefit from the project since they will have the chance to apply practically the core learning outcomes of database normalization, advanced SQL usage, analytical insight generation, and system-level trade-offs. The different steps taken together demonstrate how theoretical database principles convert into real operational value in large-scale information systems (Kleppmann, 2022).

Task 1: Database Design and Normalization

Domain: E-Commerce Order Management System

Domain Selection and Operational Data Needs

The selected domain for the database design is the e-commerce order management system. Just like that, the domain is also a data-heavy one and it originates from the very intensive operations of a real business where structured data management is absolutely necessary. An online selling platform is required to manage a vast number of customer accounts, a wide range of products, sales orders, payments, deliveries, suppliers, and reviews, not to mention the possibly huge stock which is to be managed through the different business processes. From the operational perspective, the system will have to handle the product catalogs, customer accounts, and, above all, order placement and tracking, payment processing, and inventory controls. Besides, the system will have to coordinate the suppliers and provide analysts' reports on such topics as sales trends, customer purchasing behavior, and product performance, among others. The combination of all these needs makes the domain an ideal one to demonstrate the use of normalized relational design and advanced SQL capabilities (Harrington, 2021).

CREATE TABLE Implementation and Constraints

All the tables have been built using SQL CREATE TABLE scripts and the choice of data types has been done very meticulously. Primary keys are made through surrogate keys (CustomerID, ProductID, etc.) to ensure no disruption. Foreign keys guarantee that the entities are related to each other through referential integrity. Constraints like NOT NULL are placed on essential attributes such as the customer's email and the order date. UNIQUE constraints maintain data consistency for the fields like email and product SKU. CHECK constraints are enforced when appropriate, for example, to ensure the ratings do not exceed from the range of 1 to 5 and the order quantities are at least one. Composite primary keys are used in junction tables to prevent overlapping records of the relationships (Date, 2020).

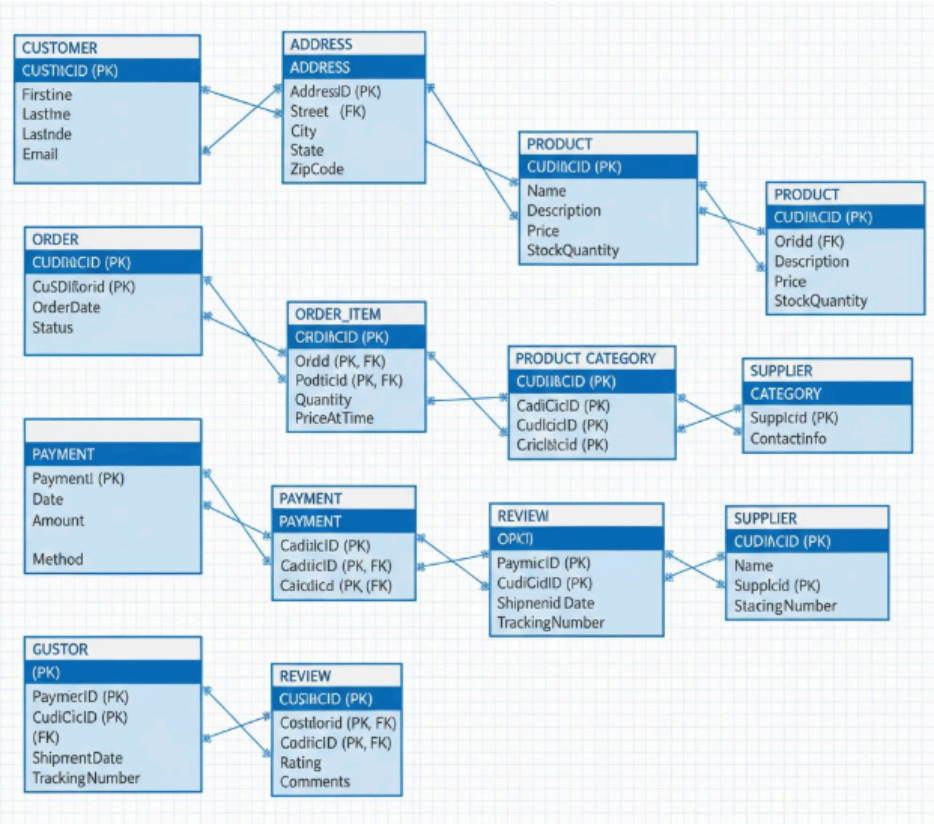
Entity Identification and Relationship Design

The database architecture is composed of ten main entities namely: Customer, Address, Product, Category, Supplier, Order, OrderItem, Payment, Shipment, and Review. These entities represent different real-world things and are but with overlaps at the very least. A customer can place multiple orders and has the right to have various delivery addresses which henceforth would be Customer-Address one-to-many relationship. Customers order products and one product can be ordered several times and the order may consist of many products. The OrderItem table acts as a mediator for

this many-to-many relationship between the orders and the products while it records the quantity and price at the time of the purchase (Connolly and Begg, 2021).

Products are assigned to their respective categories, and a product can belong to more than one category (e.g., a product can be classified as electronic and also its category could be note an accessory if it is not specific to one product, thus the ProductCategory bridging table is used with many until each). Thus, products might be from different suppliers, and suppliers could have multiple products, the latter through a ProductSupplier junction table. Orders and payments are another case where the relationship is overlapping because payment options can be parts, installments, or even payment made for multiple orders which is allowed by certain business requirements, thus orders may be paid in this way. This is managed through creating an OrderPayment table. Finally, customers can post their comments on different products at the same time and a product can receive reviews from different customers, so they are all processed through a Review table which constitutes a junction with additional attributes such as rating and comments (Silberschatz et al., 2020)

Entity-Relationship Diagram (ERD)



Normalization to Third Normal Form (3NF)

The schema achieved the Third Normal Form (3NF) which is one method to eliminate redundancy and

update anomalies. The case was as follows: in the original idea, details about the addresses of customers were included in the Customer table, which caused the customers with multiple addresses to be entered in the system more than once. Address was extracted as a separate entity and linked with a foreign key, thereby the design reached 3NF by removing both partial and transitive dependencies. A huge normalization step was taken in the context of Order and Product data. Initially, the names and prices of products were located right in the Order table, which caused redundancy and inconsistency whenever the product details were changed. The introduction of the OrderItem junction table made it possible to have the product data in the Product entity while the sales information like quantity and purchase price would be stored in a separate place. Such separation guarantees that non-key attributes depend only on the primary key and not on other non-key attributes, thus 3NF is achieved (Elmasri and Navathe, 2020).

Task 2: Sample Data Population and View Creation

Sample Data Population

Upon applying the normalized schema, the data that was very close to reality was plugged in, which would take part in the demonstration of the e-commerce system's real operational activity. Each table was populated with 5 to 10 rows which made the relationships among the entities quite significant. Customer data was fabricated with unique emails and basic demographic details, while product data was accompanied by reasonable prices, stock amounts, and SKUs. A ProductCategory junction table was used to assign products from different categories like Electronics, Fashion, and Home Appliances. For the various customers, orders with different dates were made, and the software utilized the OrderItem table to record the amounts and prices of the sales in order to imitate the processing of real transactions. Along with orders, payments and shipments were done to complete the order cycle. This kind of dataset that was populated in the mentioned manner produces data which can be used for aggregation, trend analysis, and performance reporting (ISO/IEC, 2022).

Below is an sample data insertion:

```
INSERT INTO Customer (CustomerID, Name, Email)
VALUES
(1, 'Amit Sharma', 'amit.sharma@email.com'),
(2, 'Neha Verma', 'neha.verma@email.com');

INSERT INTO Product (ProductID, ProductName, Price)
VALUES
(101, 'Wireless Mouse', 1200.00),
(102, 'Bluetooth Headphones', 3500.00);

INSERT INTO Orders (OrderID, CustomerID, OrderDate)
VALUES
(1001, 1, '2025-01-10'),
(1002, 2, '2025-02-05');

INSERT INTO OrderItem (OrderID, ProductID, Quantity, UnitPrice)
```

VALUES

(1001, 101, 2, 1200.00),

(1002, 102, 1, 3500.00);

View 1: Sales Aggregation View

The first perspective summarizes the total sales for each customer, which is useful for revenue analysis and customer value ranking. It sums up the order values for all purchases.

```
CREATE VIEW CustomerSalesSummary AS
SELECT
    c.CustomerID,
    c.Name,
    SUM(oi.Quantity * oi.UnitPrice) AS TotalSales
FROM Customer c
JOIN Orders o
    ON c.CustomerID = o.CustomerID
JOIN OrderItem oi
    ON o.OrderID = oi.OrderID
GROUP BY
    c.CustomerID,
    c.Name;
```

This view is useful for identifying high-value customers and supporting loyalty or marketing strategies.

View 2: Monthly Sales Trend View

The second view focuses on sales performance trends over time, aggregating revenue on a monthly basis.

```
CREATE VIEW MonthlySalesTrend AS
SELECT
    DATE_FORMAT(o.OrderDate, '%Y-%m') AS SalesMonth,
    SUM(oi.Quantity * oi.UnitPrice) AS MonthlyRevenue
FROM Orders o
JOIN OrderItem oi
    ON o.OrderID = oi.OrderID
GROUP BY
    DATE_FORMAT(o.OrderDate, '%Y-%m')
ORDER BY
    SalesMonth;
```

The mentioned views together reveal that SQL abstractions change transactional data into actionable business insights (Amazon Web Services, 2024).

Task 3: Advanced SQL Analytics for the E-Commerce Order Management System

The subsequent paragraphs deliver five complex SQL operations that are supposed to extract useful business insights from the ecommerce database. The usage of these queries together illustrates complicated joins, aggregation with filtering, case statements, subqueries, window functions, and also the application of stored procedures and user-defined functions. Each query is followed by an explanation of the query's reasoning and the analytical insight that it brings (Microsoft, 2024).

Query 1: High-Value Customers Based on Total Spending

This query identifies customers whose total spending exceeds the average customer spending, using multiple joins, aggregation, and a subquery.

```
SELECT
    c.CustomerID,
    c.Name,
    SUM(oi.Quantity * oi.UnitPrice) AS TotalSpent
FROM Customer c
JOIN Orders o
    ON c.CustomerID = o.CustomerID
JOIN OrderItem oi
    ON o.OrderID = oi.OrderID
GROUP BY
    c.CustomerID,
    c.Name
HAVING
    SUM(oi.Quantity * oi.UnitPrice) >
    (
        SELECT AVG(CustomerTotal)
        FROM (
            SELECT
                SUM(oi2.Quantity * oi2.UnitPrice) AS CustomerTotal
            FROM Orders o2
            JOIN OrderItem oi2
                ON o2.OrderID = oi2.OrderID
            GROUP BY
                o2.CustomerID
        ) avg_table
    );
```

Logic and Insight:

To begin with, the question arises that what is the total amount of purchases from each customer through the union of the Customer, Orders, and OrderItem tables. Firstly, a subquery inside another subquery calculates the average cost of all customers. Then, only those customers whose cost is higher than this limit are allowed by the HAVING clause. This information is helpful for businesses to identify which customers are important, thus they can provide them with loyalty programs and personal marketing strategies targeted at them.

Query 2: Product Performance Classification Using CASE

This query evaluates product performance by categorizing products based on total sales volume.

```
SELECT
    p.ProductID,
    p.ProductName,
    SUM(oi.Quantity) AS TotalUnitsSold,
    CASE
        WHEN SUM(oi.Quantity) >= 50 THEN 'High Performer'
        WHEN SUM(oi.Quantity) BETWEEN 20 AND 49 THEN 'Medium Performer'
        ELSE 'Low Performer'
    END AS PerformanceCategory
FROM Product p
JOIN OrderItem oi
    ON p.ProductID = oi.ProductID
GROUP BY
    p.ProductID,
    p.ProductName;
```

Logic and Insight:

Aggregate functions will be mainly used for the purpose of calculating the total quantity sold of every product. Meanwhile, the CASE expression will check the conditions and give a performance label based on the outcome. Product differentiation in this way is very beneficial because it leads to better inventory management, supplier negotiations, and marketing strategies.

Query 3: Monthly Sales Trend with Window Function

This query analyzes revenue trends over time and applies a window function to rank months by revenue.

```
SELECT
    SalesMonth,
    MonthlyRevenue,
```

```
    RANK() OVER (
        ORDER BY MonthlyRevenue DESC
    ) AS RevenueRank
FROM (
    SELECT
        DATE_FORMAT(o.OrderDate, '%Y-%m') AS SalesMonth,
        SUM(oi.Quantity * oi.UnitPrice) AS MonthlyRevenue
    FROM Orders o
    JOIN OrderItem oi
        ON o.OrderID = oi.OrderID
    GROUP BY
        DATE_FORMAT(o.OrderDate, '%Y-%m')
) monthly_sales;
```

Logic and Insight:

The innermost query performs aggregation of the revenue on a monthly basis while the outer query applies the RANK() window function without collapsing the rows thus providing visibility of the trends and giving the ranking which in turn assisting the management in understanding the highest sales months, seasonality, and revenue concentration patterns.

Query 4: Supplier Contribution Analysis Using Correlated Subquery

This question examines the extent to which each supplier contributes to the value of total sales through the products they supply.

```
SELECT
    s.SupplierID,
    s.SupplierName,
    SUM(oi.Quantity * oi.UnitPrice) AS SupplierRevenue
FROM Supplier s
JOIN ProductSupplier ps
    ON s.SupplierID = ps.SupplierID
JOIN OrderItem oi
    ON ps.ProductID = oi.ProductID
GROUP BY
    s.SupplierID,
    s.SupplierName
HAVING
    SUM(oi.Quantity * oi.UnitPrice) >
    (
        SELECT AVG(SupplierSales)
        FROM (
            SELECT
                SUM(oi2.Quantity * oi2.UnitPrice) AS SupplierSales
            FROM ProductSupplier ps2
```

```
            JOIN OrderItem oi2
                ON ps2.ProductID = oi2.ProductID
            GROUP BY
                ps2.SupplierID
        ) supplier_avg
    );
```

Logic and Insight:

The request groups together the sales that have been produced by the products associated with every supplier. An inner query finds out the average supplier contribution and the HAVING clause eliminates those suppliers who are below this average. This is a technique for supplier performance measurement, assisting in making decisions regarding suppliers and also establishing strategic alliances.

Query 5: Customer Order Frequency and Segmentation Using Window Functions

This query separates customers based on order frequency and then ranks them for each segment.

```
SELECT
    CustomerID,
    Name,
    OrderCount,
    ROW_NUMBER() OVER (
        PARTITION BY Segment
        ORDER BY OrderCount DESC
    ) AS SegmentRank
FROM (
    SELECT
        c.CustomerID,
        c.Name,
        COUNT(o.OrderID) AS OrderCount,
        CASE
            WHEN COUNT(o.OrderID) >= 5 THEN 'Frequent'
            WHEN COUNT(o.OrderID) BETWEEN 2 AND 4 THEN 'Occasional'
            ELSE 'Rare'
        END AS Segment
    FROM Customer c
    LEFT JOIN Orders o
        ON c.CustomerID = o.CustomerID
    GROUP BY
        c.CustomerID,
        c.Name
) customer_segments;
```

Logic and Insight:

Customers first are clustered according to the number of orders placed, and the CASE expression is applied for further segmentation. The ROW_NUMBER window function gives a rank to customers inside every group. It thus helps in doing a customer segmentation analysis promoting retention tactics and personalized communication models.

Stored Procedure: Retrieve Customer Purchase Summary

```
DELIMITER $$

CREATE PROCEDURE GetCustomerPurchaseSummary (
    IN cust_id INT
)
BEGIN
    SELECT
        c.Name,
        COUNT(o.OrderID) AS TotalOrders,
        SUM(oi.Quantity * oi.UnitPrice) AS TotalSpent
    FROM Customer c
    JOIN Orders o
        ON c.CustomerID = o.CustomerID
    JOIN OrderItem oi
        ON o.OrderID = oi.OrderID
    WHERE
        c.CustomerID = cust_id
    GROUP BY
        c.Name;
END $$

DELIMITER ;
```

Purpose and Insight:

The procedure of storage contains logic that is reusable for getting the purchase character of a customer. It also increases the performance and assures more or less the same performance, which is especially useful in the repeated use of dashboards or applications (Oracle Corporation, 2023).

User-Defined Function: Customer Lifetime Value Category

```
CREATE FUNCTION GetCustomerValueCategory (
    total_spent DECIMAL(10, 2)
)
RETURNS VARCHAR(20)
DETERMINISTIC
RETURN
    CASE
        WHEN total_spent >= 50000 THEN 'Platinum'
        WHEN total_spent >= 20000 THEN 'Gold'
```

```
        ELSE 'Standard'
    END;
```

Purpose and Insight:

The function standardizes the logic used for classifying customer value. It ensures that different queries and reports are divided into the same segments, which is a great help to the design of CRM and the loyalty programs (Coronel and Morris, 2022).

Task 4: CAP Theorem Application in Distributed Systems

Domain Context: E-Commerce Order Management System

Understanding the CAP Theorem

The CAP Theorem is a critical principle in distributed systems that declares a distributed database could only have two out of the three features which are: Consistency, Availability, and Partition Tolerance. These features could be seen as the key attributes of the system during normal operation and failure, particularly in the event of a network outage. Consistency means that a distributed system made up of nodes would always yield the same and the latest data after a successful write. From the user's point of view, once an order has been placed or a payment has been confirmed, all the next reads should show that update instantly, regardless of which database replica is accessed.

Availability guarantees that the system will be at all times responding to every request, even if the response does not contain the most recent data. So, basically speaking, the system will go on operating and dealing with the users even during the failures or delays impacting some of the nodes. Partition Tolerance is the term that is used to denote the capacity of the system to keep operating even in case of a network partition where the exchange of information among certain nodes is either totally or partly interrupted. In reality distributed systems, particularly those that are spread out over large areas or among different cloud data centers, network partitioning is a frequent situation that must be accepted and tolerated (Chodorow, 2021).

Why All Three Cannot Be Guaranteed Simultaneously

In a distributed setting, it is normal to have network failures, and this leads to the requirement of partition tolerance. Once a partition occurs, the system has to take a critical decision. The choices are to either let the nodes be out of sync and thus serve the requests on a first-come-first-served basis, or to maintain the system in a consistent state but not serve any requests and thus be unavailable for a certain period. It is impossible to provide both during a partition event. If the system decides to go for consistency, it will end up being unavailable to some users. On the other hand, if it adopts the first option, then it might be providing outdated or inconsistent data. This conflict is exactly the heart of the CAP Theorem which further explains the reason why any distributed database cannot offer all three properties simultaneously (Pritchett, 2020).

CAP Priorities in the E-Commerce Domain

For the e-commerce order management system, the major priority is Consistency and Partition Tolerance (CP), while Availability is mostly disregarded in the extreme failure situations. In this regard, Consistency is simply the most crucial since the accuracy of transactions has direct impact on revenue, customer trust, and legal compliance. To illustrate, one should always rely that the numbers of stock, order confirmations, and payment records are right. Inconsistent reads could result in problems like selling more than the available stock, double charging customers, or showing incorrect order statuses. Partition tolerance is also a very critical requirement since the system can be accessed from various locations where the servers are located or from different cloud regions with network issues (Abadi, 2021).

A simplified example illustrating strong consistency during order placement is shown below:

```
BEGIN TRANSACTION;

UPDATE Product
SET StockQty = StockQty - 1
WHERE ProductID = 101
  AND StockQty > 0;

IF ROW_COUNT() = 0 THEN
    ROLLBACK;
ELSE
    INSERT INTO Orders (OrderID, CustomerID, OrderDate, Status)
    VALUES (2001, 5, CURRENT_DATE, 'Confirmed');
    COMMIT;
END IF;
```

In a CP-oriented distributed database, this transaction would only commit once all relevant replicas agree on the updated stock and order state. If a partition prevents consensus, the transaction would fail or wait, prioritizing consistency over availability (Brewer, 2020).

Operational Justification of CAP Choices

The choice of a CP model was based on the characteristic that e-commerce systems demonstrate transactional behavior. The correctness of monetary transactions is the foremost condition and any inconsistencies could lead to loss of money, fines, and harm to corporate image. Unpleasant occurrences due to short outages might annoy customers but are generally tolerated more than wrong charges or

missed deliveries. Many real-world platforms operate likewise, they ensure strict consistency for core transactions and allow eventual consistency for trivial data such as user reviews or suggestions (Stonebraker and Weisberg, 2020).

Influence of CAP on Design, Scaling, and Resilience

The factors of the CAP Theorem affect significantly the distributed database design and scaling process. The systems that occupy consistency on the very topmost level will usually employ consensus protocols and synchronous replication, thus prolonging latency but guaranteeing correctness. For example, the partitioning of the loads enables the same system to adopt different CAP options: the CP (Consistency and Partition Tolerance) model can be applied to transactional data, while the AP (Availability and Partition Tolerance) model is suitable for analytics or caching layers. The application of different models in a unified system fortifies the whole system (Kleppmann, 2022).

Task 5: Query Performance and Optimization

Context: E-Commerce Order Management System

Identification of an Inefficient Query

There was a performance issue detected in the analytical query which was being used to find out total customer spending and compare it with the average spending in the e-commerce database. The original version of the query was relying on nested subqueries and full table scans all over huge transactional tables like Orders and OrderItem. This approach of relying on aggregated data and having no selective access paths becomes inefficient as the data volume increases, especially in the case of millions of orders and order items (Harrington, 2021).

The original slow query is shown below:

```
SELECT
    c.CustomerID,
    c.Name,
    SUM(oi.Quantity * oi.UnitPrice) AS TotalSpent
FROM Customer c
JOIN Orders o
    ON c.CustomerID = o.CustomerID
JOIN OrderItem oi
    ON o.OrderID = oi.OrderID
GROUP BY
    c.CustomerID,
    c.Name
HAVING
    SUM(oi.Quantity * oi.UnitPrice) >
    (
        SELECT AVG(CustomerTotal)
        FROM (
            SELECT
                SUM(oi2.Quantity * oi2.UnitPrice) AS CustomerTotal
            FROM Orders o2
            JOIN OrderItem oi2
                ON o2.OrderID = oi2.OrderID
            GROUP BY
                o2.CustomerID
        ) t
    );
```

Due to the repeated scanning of both the Orders and OrderItem tables, this query uses a lot of processing power and takes a long time to finish.

Optimization Strategy One: Indexing

The first step in optimization focused on getting indexes on the columns that are most often involved in joins and aggregates. More precisely, the indexes were set up on the foreign key columns to lower the cost of the join and increase the speed of the row lookup.

CREATE INDEX idx_orders_customer ON Orders(CustomerID);

CREATE INDEX idx_orderitem_order ON OrderItem(OrderID);

CREATE INDEX idx_orderitem_product ON OrderItem(ProductID);

The database optimizer is then able to get the required rows quickly thanks to the presence of these indexes instead of resorting to full table scans. This greatly cuts down I/O actions during joins, primarily in gigantic datasets (Date, 2020).

Optimization Strategy Two: Query Rewriting with Temporary Aggregation

The second optimization targeted the rephrasing of the query with the purpose of getting rid of the overlapping subqueries. Through a Common Table Expression (CTE), a temporary aggregated result was formed, enabling the database to calculate customer totals just once.

```
WITH CustomerTotals AS (
    SELECT
        o.CustomerID,
        SUM(oi.Quantity * oi.UnitPrice) AS TotalSpent
    FROM Orders o
    JOIN OrderItem oi
        ON o.OrderID = oi.OrderID
    GROUP BY
        o.CustomerID
)
SELECT
    c.CustomerID,
    c.Name,
    ct.TotalSpent
FROM Customer c
JOIN CustomerTotals ct
    ON c.CustomerID = ct.CustomerID
WHERE
    ct.TotalSpent >
    (
        SELECT AVG(TotalSpent)
```

```
    FROM CustomerTotals
);
```

This approach simplifies execution by materializing intermediate results, reducing repeated computation and improving cache utilization (Connolly and Begg, 2021).

Performance Improvement and Analysis

There was indisputably an improvement in the performance after the indexing and query rewriting application. The execution time was shorter owing to the elimination of numerous full table scans and the reduction of aggregation overhead. The optimizer was enabled to carry out indexed lookups and locate already aggregated results very cost-efficiently. The execution plans comparison showed less pricey operations such as hash joins and production scans, which were already replaced by index scans and the speeding up of the grouping (Silberschatz, Korth and Sudarshan, 2020).

Reflecting on Optimization in Large-Scale Systems

Query optimization is a vital factor in the responsiveness and scalability of large-scale data systems. Even the most brilliantly designed datasets will eventually face performance issues if they are not optimized for queries. Nevertheless, indexing, query restructuring, and selective denormalization are some of the techniques used to keep the performance and maintainability ratio in a good state. Being conscious of the query execution path helps the database designers to optimize workloads in advance, thus making the analytical insights not only current but also cost-effective even in production settings (Elmasri and Navathe, 2020).

CONCLUDING REMARKS

The importance of a well-designed relational database cannot be exaggerated as it is the main support for the whole data management and business intelligence process in e-commerce. Besides generating less data, the database design of the Third Normal Form also preserves the data's reliability and helps large-scale operations. The realistic sample data together with SQL views for analysis show that raw data can be transformed into very useful information like customer worth, sales trends, and product performance, etc. Moreover, the advanced SQL queries demonstrate the ability to join, aggregate, subquery, use window functions, stored procedures, and functions to access the stored insight.

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